

Dhruv Gupta

Engineering Specialist · Agentic AI, Multi-Agent Systems & Python Backend

Gurugram, India • +91 99967 54084 • dhruvgupta9814@gmail.com • linkedin.com/in/dhruvgupta14

PROFESSIONAL SUMMARY

Senior AI backend engineer with 5 years building and shipping production-grade Agentic AI systems and Python services. Hands-on developer of autonomous and semi-autonomous multi-agent applications using LangGraph, LangChain, MCP (Model Context Protocol) servers, and the A2A (agent-to-agent) protocol. Delivered real-time, production-deployed platforms driving 40-70% efficiency gains, 25% cost reduction, and 5x revenue growth across enterprise client engagements.

TECHNICAL SKILLS

Agentic AI & Orchestration: Multi-agent systems, autonomous & semi-autonomous agents, MCP (Model Context Protocol) servers, A2A (agent-to-agent) protocol, tool / function calling, LangGraph, LangChain

LLM & GenAI: RAG pipelines, corrective RAG, prompt engineering, structured data extraction, entity recognition, vector databases, OpenAI, Claude (Anthropic)

LLM Observability: Langfuse

Backend & APIs: Python, FastAPI, Flask, RESTful APIs, Celery, PostgreSQL, OpenSearch

Cloud & DevOps: AWS (EC2, EKS, Lambda, S3, RDS, Aurora, Neptune), Docker, CI/CD, Git

Frontend & Other: React, JavaScript, Agile / Scrum, performance tuning, unit testing, team leadership

PROFESSIONAL EXPERIENCE

Engineering Specialist · ZS Associates · Gurugram, India *Dec 2025 - Present*

- **Medical Content Authoring Platform:** Architected and shipped a production Agentic AI system that autonomously drafts regulated clinical documents, cutting medical-writer effort by 60-70%.
- Designed the multi-agent orchestration layer with LangGraph and LangChain, integrating MCP servers for tool access and the A2A protocol for agent-to-agent coordination across an end-to-end document digitization and authoring pipeline.
- Personally owned agent-workflow design, MCP tool integration, and the FastAPI backend services powering the platform.

Senior Engineer · ZS Associates · Gurugram, India *Jul 2023 - Dec 2025*

- **Trial Synthesizer (Agentic AI):** Built an autonomous feasibility agent that mines historical and open-source clinical-trial data to surface planning insights, cutting manual analysis effort by 75% and saving ~600-800 hours annually. Implemented agentic workflows with LangGraph, LangChain, FastAPI, and PostgreSQL.
- **IDP Generator (Agentic AI):** Engineered a multi-step generative agent that auto-produces integrated drug-development roadmaps spanning timelines, costs, and cross-team dependencies, reducing planning effort by ~80%. Built with OpenAI, AWS Neptune (graph DB), FastAPI, and PostgreSQL.
- **GenAI & RAG Systems:** Architected and deployed production RAG chatbots with corrective-RAG pipelines across PDF, DOCX, and PPT sources; automated entity recognition and structured data extraction via prompt engineering.
- **Internal Tools & Platforms:** Built FastAPI / PostgreSQL backends and web portals with SSO authentication and in-app chatbot assistance, reducing project approval and management time by 50%.
- **Leadership:** Led a cross-functional team of 10+ engineers, owning sprint planning, deployments, and onboarding of new members.

Software Developer · ZS Associates · Gurugram, India *Jul 2021 - Jul 2023*

- **Strategic System Migration:** Migrated a \$7M client's forecasting solution from offline Excel to a real-time web platform (React, JavaScript, OfficeJS), directly contributing to 5x revenue growth through faster, data-driven decisions.
- Built and scaled multiple client-facing and internal tools (React, FastAPI, PostgreSQL), improving response times and supporting a growing user base.

ACHIEVEMENTS

- **Project Champion Award** - for delivering a GenAI-based solution on a client project.
- **Innovation Excellence Project Award (2022)** - for transitioning a legacy forecasting solution to a modern platform.

EDUCATION

B.Tech, Electronics & Communication Engineering *CGPA: 8.7*

The LNM Institute of Information Technology

Swami Vivekanand Public School - Class XII: 96% | Class X: 95%